Complete Beginner's Guide to Setting Up Your Cape Town Microgreens Business

Author: Manus AI
Date: July 2025
For: Complete Beginners with 18 Vertical Pipes and 2x1000L Tanks

IMPORTANT: This Guide is for Complete Beginners

This guide assumes you have ZERO experience with hydroponics or microgreens. Every step is explained in detail with exact measurements, specific products to buy, and where to buy them in Cape Town. Follow this guide step-by-step and you will have a working microgreens business.

What You Will Achieve

By following this guide exactly, you will:

- Transform your 18 pipes and 2 tanks into a profitable microgreens system
- Start selling microgreens within 30 days
- Generate R15,000-25,000 monthly income within 90 days
- Have a complete understanding of microgreens production

Chapter 1: Understanding Your Current Setup

What You Already Have

You mentioned you have 18 vertical pipes and 2x1000-liter tanks. Before we start, let's understand exactly what this means and how we'll use it.

Your 18 Vertical Pipes: These are likely PVC pipes arranged vertically with holes cut in them for plants. Each pipe can hold approximately 20-30 plant sites (holes). This gives you a total capacity of 360-540 growing sites. For microgreens, this is excellent because microgreens are small and grow quickly.

Your 2x1000-Liter Tanks: These tanks will serve different purposes:

- Tank 1: Clean water and nutrient solution storage
- Tank 2: Used water collection and recycling

What This Setup Can Produce: With proper management, your system can produce:

- 15-25 kg of microgreens per month initially
- 35-50 kg per month once optimized
- Monthly revenue of R15,000-60,000 depending on varieties and customers

Understanding Microgreens vs Regular Plants

What Are Microgreens? Microgreens are vegetables and herbs harvested when they are 7-21 days old, just after the first true leaves appear. They are:

- Much smaller than full-grown plants
- Packed with nutrients (up to 40x more than mature plants)
- Quick to grow (7-14 days from seed to harvest)
- Expensive to buy (R100-150 per 100g in Cape Town)
- Perfect for your vertical system

Why Microgreens Are Perfect for Beginners:

- Fast results (you see success in 1-2 weeks)
- High profit margins
- Small space requirements
- Less can go wrong compared to full-size plants
- Quick learning cycle

Chapter 2: Essential Equipment You Need to Buy

Critical Equipment Shopping List

Before you can start, you need to buy specific equipment. Here’s exactly what you need and where to get it in Cape Town:

WATER PUMPS (Essential - Buy First)

- What to buy: 2x submersible water pumps, 3000L/hour capacity
- Why you need it: To move water from tanks to pipes and back
- Where to buy: Builders Warehouse, Makro, or Hydroponic.co.za
- Approximate cost: R800-1200 each
- Specific model to look for: “Aquarium submersible pump 3000L/h” or “Pond pump 3000L/h”

WATER TUBING AND FITTINGS

- What to buy: 50 meters of 25mm flexible tubing + fittings
- Why you need it: To connect pumps to pipes
- Where to buy: Builders Warehouse plumbing section
- Approximate cost: R15-20 per meter + R200 for fittings
- Ask for: “25mm flexible irrigation tubing and T-joints”

GROWING MEDIUM

- What to buy: Coconut coir (coco peat) - 10x compressed blocks
- Why you need it: Microgreens need something to grow in
- Where to buy: Garden centers, Stodels, or online
- Approximate cost: R50-80 per block
- Alternative: Perlite mixed with vermiculite (50⁄50 mix)

SEEDS (Start with These 4 Varieties)

- Broccoli microgreen seeds - 500g
- Radish microgreen seeds - 500g

- Pea shoot seeds - 1kg
- Sunflower microgreen seeds - 1kg
- Where to buy: Livingseeds.co.za, Eden Seeds, or Starke Ayres
- Approximate cost: R150-300 per variety

NUTRIENTS

- What to buy: Hydroponic nutrient solution (A+B formula)
- Specific product: “Hydroponic Nutrients A+B 1L each”
- Where to buy: Hydroponic.co.za or garden centers
- Approximate cost: R300-500 for starter kit
- Alternative: “Nutrifeed” or “Culterra” hydroponic nutrients

pH TESTING KIT

- What to buy: Digital pH meter or pH test strips
- Why you need it: Plants need specific pH levels (5.5-6.5)
- Where to buy: Pool shops, hydroponic stores, or online
- Approximate cost: R200-500
- Easier option: pH test strips (R50-100)

GROWING TRAYS

- What to buy: 50x seedling trays (the black plastic ones)
- Size: Standard seedling tray size (about 35cm x 25cm)
- Where to buy: Garden centers, Builders Warehouse
- Approximate cost: R15-25 each
- Alternative: Use clean margarine containers with holes

BASIC TOOLS

- Sharp knife or scissors for harvesting
- Measuring cups (1L and 500ml)
- Spray bottle for misting
- Thermometer

- Timer or phone for scheduling

Total Initial Investment

Essential Equipment: R4,000-6,000 Seeds and Nutrients: R1,500-2,500 Growing Supplies: R1,000-1,500 TOTAL: R6,500-10,000

This is much less than the R180,000 I mentioned in the strategy documents because we're starting simple and building up.

Chapter 3: Step-by-Step System Setup

Week 1: Basic System Assembly

Day 1: Assess and Clean Your Setup

Morning (2 hours):

1. Take photos of your current 18 pipes and 2 tanks
2. Measure the distance between tanks and pipes
3. Check if your pipes have drainage holes at the bottom
4. Clean both tanks thoroughly with bleach solution (1:10 ratio)
5. Rinse tanks completely - no bleach smell should remain

Afternoon (2 hours):

1. Test that your pipes are stable and won't fall over
2. Check that water can flow from top to bottom of each pipe
3. Identify where you'll place pumps (in Tank 1)
4. Plan your tubing route from Tank 1 to pipes
5. Make a shopping list based on what you found

Day 2: Equipment Shopping

Morning (3 hours):

1. Go to Builders Warehouse or Makro

2. Buy pumps, tubing, and fittings first
3. Test pumps in store if possible (ask staff)
4. Buy pH testing kit
5. Get basic tools if you don't have them

Afternoon (2 hours):

1. Visit garden center for growing medium and trays
2. Order seeds online or buy locally
3. Buy nutrients from hydroponic store
4. Get spray bottle and measuring cups

Day 3: Install Water System

Morning (3 hours):

1. Install first pump in Tank 1
2. Connect tubing from pump to your pipes
3. Create a distribution system so water goes to all 18 pipes
4. Test water flow - water should reach all pipes

Afternoon (2 hours):

1. Install drainage system from pipes back to Tank 2
2. Install second pump in Tank 2 to return water to Tank 1
3. Test complete water cycle
4. Fix any leaks or flow problems

Day 4: Test Your Water System

Morning (2 hours):

1. Fill Tank 1 with clean water
2. Turn on pumps and run system for 30 minutes
3. Check that water flows to all pipes evenly
4. Verify water returns to Tank 2

5. Check for leaks and fix them

Afternoon (2 hours):

1. Run system for 2 hours continuously
2. Measure how much water you lose to evaporation
3. Adjust pump timing if needed
4. Clean any debris from system

Day 5: Prepare Growing Medium

Morning (2 hours):

1. Soak coconut coir blocks in water (they expand 5x)
2. Break up expanded coir until fluffy
3. Fill 20 growing trays with coir
4. Moisten coir but don't make it soggy

Afternoon (1 hour):

1. Place filled trays in your pipe system
2. Test that trays fit properly
3. Ensure water can reach the coir in each tray

Day 6: Mix Your First Nutrient Solution

Morning (1 hour):

1. Read nutrient instructions carefully
2. Mix nutrients in Tank 1 according to package directions
3. Usually: 5ml of Part A + 5ml of Part B per liter of water
4. Test pH - should be 5.5-6.5
5. Adjust pH if needed (add lemon juice to lower, baking soda to raise)

Afternoon (1 hour):

1. Run nutrient solution through system
2. Let it circulate for 1 hour

3. Test pH again - it may change
4. Take notes on what you did

Day 7: Final System Test

Morning (2 hours):

1. Run complete system with nutrients for 3 hours
2. Check all pipes get nutrient solution
3. Verify drainage works properly
4. Test pH one more time

Afternoon (1 hour):

1. Clean and organize your workspace
2. Prepare for planting tomorrow
3. Review seed planting instructions
4. Set up your record-keeping system

Week 2: First Planting

Day 8: Plant Your First Microgreens

Morning (2 hours):

1. Start with broccoli - it's the easiest
2. Measure 2 tablespoons of broccoli seeds
3. Soak seeds in water for 2 hours
4. Spread soaked seeds evenly on 4 trays of moist coir
5. Cover seeds lightly with more coir
6. Place trays in your pipe system

Afternoon (1 hour):

1. Turn on water system
2. Set timer to run pumps 15 minutes every 2 hours
3. Mist seed trays lightly with spray bottle

4. Cover trays with newspaper to keep dark
5. Record what you did and when

Day 9: Daily Monitoring Begins

Morning (15 minutes):

1. Check if seeds have started sprouting
2. Remove newspaper when you see green shoots
3. Mist trays if coir looks dry
4. Check water level in tanks

Evening (15 minutes):

1. Check sprouts again
2. Adjust pump timing if needed
3. Record observations

Day 10-14: Watch Your First Crop Grow

Daily routine (30 minutes total): Morning (15 minutes):

1. Check sprout growth
2. Mist if needed
3. Check water and nutrient levels
4. Take photos to track progress

Evening (15 minutes):

1. Check for any problems
2. Adjust lighting if sprouts look pale
3. Record daily observations

Day 15: Your First Harvest!

Morning (1 hour):

1. Your broccoli microgreens should be 3-5cm tall
2. Cut them just above the coir level with sharp scissors

3. Rinse harvested microgreens in cold water
4. Dry gently with paper towels
5. Weigh your harvest

Afternoon (1 hour):

1. Package microgreens in small containers
2. Calculate your yield per tray
3. Clean empty trays for replanting
4. Celebrate - you just grew your first crop!

Chapter 4: Scaling Up Production

Week 3: Plant More Varieties

Now that you've successfully grown broccoli, it's time to expand:

Day 16: Plant Radish Microgreens

- Follow same process as broccoli
- Radish grows faster (7-10 days)
- Use 6 trays this time

Day 18: Plant Pea Shoots

- Soak pea seeds overnight first
- Use larger seeds spacing
- Takes 10-14 days to harvest

Day 20: Plant Sunflower Microgreens

- Remove hulls from seeds first
- Needs more water than other varieties
- Takes 8-12 days

Week 4: Establish Production Schedule

By now you should have:

- Broccoli ready to harvest every 2 weeks
- Radish ready every 10 days
- Pea shoots every 2 weeks
- Sunflower every 10 days

Create a Planting Calendar:

- Monday: Plant new broccoli
- Wednesday: Plant new radish
- Friday: Plant new pea shoots
- Sunday: Plant new sunflower

This gives you something to harvest almost every day once established.

Chapter 5: Quality Control and Harvesting

Perfect Harvesting Technique

When to Harvest:

- Broccoli: When first true leaves appear (day 10-14)
- Radish: When cotyledons are fully open (day 7-10)
- Pea shoots: When 5-8cm tall with tendrils (day 10-14)
- Sunflower: When cotyledons are large and green (day 8-12)

How to Harvest:

1. Use sharp, clean scissors
2. Cut just above the growing medium
3. Don’t pull - this damages roots and medium
4. Harvest in early morning when plants are crisp

5. Work quickly to maintain freshness

Post-Harvest Handling:

1. Rinse gently in cold water
2. Spin dry in salad spinner or pat with paper towels
3. Package immediately in breathable containers
4. Refrigerate at 2-4°C
5. Use within 5-7 days

Quality Standards

Grade A (Premium - R120-150 per 100g):

- Uniform size and color
- No yellowing or wilting
- Clean, no growing medium attached
- Fresh, crisp texture
- No pest damage

Grade B (Standard - R80-100 per 100g):

- Slight size variation acceptable
- Minor color variation
- Still fresh and crisp
- Minimal growing medium

Grade C (Discount - R50-70 per 100g):

- Size and color variation
- Some wilting acceptable
- Still safe to eat
- Use for juice bars or processing

Chapter 6: Your First Sales

Week 5: Finding Your First Customers

Day 29: Prepare for Sales

Morning (2 hours):

1. Package your best microgreens in clear containers
2. Create simple labels with your name and phone number
3. Take high-quality photos of your products
4. Calculate your costs and set prices

Afternoon (2 hours):

1. Visit 3 nearby restaurants during slow hours (2-4 PM)
2. Ask to speak with the head chef
3. Offer free samples
4. Explain that you're a local grower
5. Leave your contact information

Day 30: Make Your First Sale

Morning (1 hour):

1. Follow up with restaurants from yesterday
2. Offer to supply them weekly
3. Start with small quantities (500g-1kg)
4. Agree on delivery schedule

Afternoon (2 hours):

1. Visit local health food stores
2. Offer to supply them on consignment
3. Provide point-of-sale materials
4. Set up regular delivery schedule

Pricing Strategy for Beginners

Restaurant Sales (Wholesale):

- Broccoli: R80 per 100g
- Radish: R70 per 100g
- Pea shoots: R90 per 100g
- Sunflower: R75 per 100g

Retail Sales (Direct to Consumer):

- Broccoli: R120 per 100g
- Radish: R100 per 100g
- Pea shoots: R130 per 100g
- Sunflower: R110 per 100g

Farmers Market Sales:

- Premium pricing (R120-150 per 100g)
- Offer mixed packs
- Provide recipe cards
- Build customer relationships

Chapter 7: Troubleshooting Common Problems

Problem 1: Seeds Not Germinating

Symptoms:

- Seeds planted 3+ days ago with no sprouting
- Seeds look moldy or smell bad
- Uneven germination

Causes and Solutions:

- **Old seeds:** Buy fresh seeds from reputable suppliers

- **Wrong temperature:** Keep growing area 18-24°C
- **Too wet/too dry:** Coir should be moist but not soggy
- **Poor quality seeds:** Test germination rate before large plantings

Prevention:

- Test small batches of new seed lots
- Store seeds in cool, dry place
- Use seeds within 2 years of purchase
- Keep detailed records of seed performance

Problem 2: Yellowing or Pale Microgreens

Symptoms:

- Microgreens look yellow instead of green
- Weak, spindly growth
- Poor flavor

Causes and Solutions:

- **Insufficient light:** Add LED grow lights or move to brighter location
- **Nutrient deficiency:** Check nutrient solution strength and pH
- **Overcrowding:** Use fewer seeds per tray
- **Poor air circulation:** Add small fan for air movement

Prevention:

- Provide 12-16 hours of light daily
- Maintain proper nutrient levels
- Follow recommended seeding rates
- Ensure good ventilation

Problem 3: Mold or Fungal Growth

Symptoms:

- White, fuzzy growth on seeds or growing medium
- Bad smell from trays
- Seeds rotting instead of sprouting

Causes and Solutions:

- **Too much moisture:** Reduce watering frequency
- **Poor air circulation:** Add ventilation
- **Contaminated growing medium:** Use fresh, sterile medium
- **High humidity:** Reduce humidity with ventilation

Prevention:

- Don't overwater
- Ensure good drainage
- Use clean equipment and containers
- Maintain proper air circulation

Problem 4: Slow Growth

Symptoms:

- Microgreens taking longer than expected to reach harvest size
- Weak, thin stems
- Poor yield

Causes and Solutions:

- **Low temperature:** Maintain 18-24°C growing temperature
- **Insufficient nutrients:** Check nutrient solution concentration
- **Wrong pH:** Maintain pH 5.5-6.5
- **Poor light:** Provide adequate lighting

Prevention:

- Monitor temperature daily
- Test and adjust nutrient solution weekly

- Check pH regularly
- Provide consistent lighting schedule

Problem 5: Poor Taste or Texture

Symptoms:

- Bitter or off flavors
- Tough, fibrous texture
- Customers complaining about quality

Causes and Solutions:

- **Harvesting too late:** Harvest at proper stage
- **Stress conditions:** Maintain optimal growing conditions
- **Poor post-harvest handling:** Improve harvesting and storage
- **Wrong varieties:** Choose varieties suited to your market

Prevention:

- Learn proper harvest timing for each variety
- Maintain consistent growing conditions
- Handle harvested microgreens gently
- Store at proper temperature

Chapter 8: Record Keeping and Business Management

Essential Records to Keep

Daily Production Log:

- Date planted
- Variety and quantity of seeds
- Growing conditions (temperature, humidity)
- Watering schedule

- Observations and problems

Harvest Records:

- Date harvested
- Variety and weight harvested
- Quality grade
- Yield per tray
- Customer destination

Financial Records:

- Daily sales and revenue
- Expenses (seeds, nutrients, packaging)
- Customer payments
- Profit calculations

Customer Records:

- Customer contact information
- Order history and preferences
- Payment terms
- Delivery schedules

Simple Business Management

Weekly Tasks:

- Calculate total production and sales
- Order supplies for next week
- Plan planting schedule
- Follow up with customers

Monthly Tasks:

- Analyze profitability by variety
- Review customer satisfaction

- Plan capacity expansion
- Update pricing if needed

Quarterly Tasks:

- Evaluate overall business performance
- Consider new varieties or markets
- Plan equipment upgrades
- Set goals for next quarter

Chapter 9: Expanding Your Business

Month 2: Increase Production

Once you're comfortable with the basics:

Week 5-6: Optimize Current System

- Fine-tune nutrient solutions
- Perfect harvest timing
- Improve packaging and presentation
- Build customer relationships

Week 7-8: Add New Varieties

- Try specialty varieties (purple radish, mustard greens)
- Test market demand for new products
- Adjust production based on sales data

Month 3: Market Expansion

Week 9-10: Find New Customers

- Approach more restaurants
- Try farmers markets
- Develop online sales

- Create subscription services

Week 11-12: Improve Efficiency

- Automate watering systems
- Streamline harvesting process
- Optimize packaging and delivery
- Reduce waste and costs

Month 4 and Beyond: Scale Up

Consider These Expansions:

- Add more growing pipes
- Install automated systems
- Hire part-time help
- Develop value-added products

Financial Milestones:

- Month 1: R5,000-10,000 revenue
- Month 2: R10,000-20,000 revenue
- Month 3: R15,000-30,000 revenue
- Month 6: R25,000-50,000 revenue

Chapter 10: Success Metrics and Goals

Key Performance Indicators

Production Metrics:

- Germination rate (target: >90%)
- Days to harvest (track by variety)
- Yield per tray (grams harvested)
- Quality grade percentage

Financial Metrics:

- Revenue per week
- Profit margin per variety
- Customer acquisition cost
- Customer lifetime value

Operational Metrics:

- Time spent on daily tasks
- Waste percentage
- Customer satisfaction scores
- On-time delivery rate

90-Day Success Plan

Days 1-30: Foundation

- Set up system correctly
- Grow first successful crops
- Find first customers
- Establish routines

Days 31-60: Growth

- Increase production volume
- Add new varieties
- Expand customer base
- Improve efficiency

Days 61-90: Optimization

- Fine-tune all processes
- Maximize profitability
- Plan for expansion
- Build sustainable business

Your Success Checklist

By Day 90, you should have:

- [ ] Consistent production of 4+ varieties
- [ ] 5-10 regular customers
- [ ] R15,000+ monthly revenue
- [ ] Efficient daily routines
- [ ] Quality control systems
- [ ] Basic business records
- [ ] Plan for expansion

Conclusion: Your Path to Success

This guide gives you everything you need to transform your 18 pipes and 2 tanks into a profitable microgreens business. The key to success is:

1. **Start Simple:** Follow this guide exactly, don't try to do everything at once
2. **Be Consistent:** Stick to your daily routines and schedules
3. **Focus on Quality:** Better to grow less with high quality than more with poor quality
4. **Learn Continuously:** Keep detailed records and learn from every crop
5. **Build Relationships:** Your customers are your most valuable asset

Remember: You don't need to be perfect from day one. Every expert was once a beginner. Follow this guide step-by-step, and within 90 days you'll have a thriving microgreens business that generates R15,000-30,000 monthly income.

Your journey from complete beginner to successful microgreens producer starts with Day 1 of this guide. Take it one day at a time, and you will succeed.